Dataset : loan Approval prediction

Problem : predict whether a person is eligible for a loan

classes :

- 0 -> loan Not Approved
- 1 -> loan Approved

input features :

- Age
- income
- Credit score
- Experiance

Fast Row :

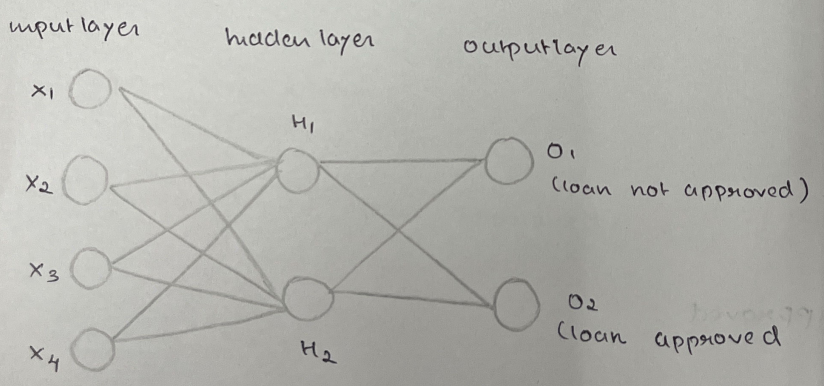
x = [30, 50, 700, 5]

(Age, income, Credit score, experiance)

Actual output = loan Approved (1)

Neural Network

input (4) -> hidden layer (2) -> output (2)



Hidden neuron H1 weight

- $w_1 = 0.01$
- $w_2 = 0.002$
- $w_3 = 0.003$
- $w_4 = 0.005$
- $B1 = 0.2$

hidden neuron H2 weight

- $w_2 = 0.005$
- $w_6 = 0.003$
- $w_7 = 0.002$
- $w_8 = 0.004$
- $B_2 = 0.1$

hidden layer formula (H_1):

$z_1 = x_1w_1 + x_2w_2 + x_3w_3 + x_4w_4 + B_1$

$z_1 = (30 \times 0.01) + (50 \times 0.002) + (700 \times 0.003) + (5 \times 0.004) + 0.2$

$z_1 = 0.30 + 0.10 + 2.10 + 0.02 + 0.2$

$= \underline{\underline{2.72}}$

Sigmoid :

$H_1 = 1/(1 + e^{-2.72})$

$H_1 \approx 0.9382$

hidden layer H_2

$z_2 = (30 \times 0.005) + (50 \times 0.003) + (700 \times 0.002) + (5 \times 0.004) + 0.1$

$= 0.15 + 0.15 + 1.40 + 0.02 + 0.1$

$z_2 = 1.82$

$H_2 = 1/1 + e^{-1.82})$

$H_2 = 0.8606$

output layer

O_1 -> loan not approved

weights :

- $w(H_1, O_1) = 0.4$
- $w(H_2, O_1) = 0.2$
- $B = 0.1$

$z_O_1 = (0.9382 \cdot 0.4) + (0.8606 \times 0.2) + 0.1$

$= 0.3753 + 0.1721 + 0.1$

$z_o1 = 0.6474$

Sigmoid :

$o1 \approx 0.6564$

$o2 \rightarrow$ loan Approved

weights :

- $W(H1, o2) = 0.3$
- $W(H2, o2) = 0.8$
- $B = 0.1$

$z_o2 = (0.9382 \cdot 0.3) + (0.8606 \times 8) + 0.1$

$= 0.2815 + 0.6885 + 0.1$

Sigmoid :

$o2 \approx 0.7446$

output value

$[0.6564 \quad 0.7446]$

Since :

$0.7446 > 0.6564$

Final prediction = loan Approved